

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: 3/30/2020 1:47:25 PM
To: "Sneller, Cindy (EGLE)" <SNELLERC@michigan.gov>
Subject: RMP Sludge Data
Attachments: TVS Sludge.xlsx, Lab Data Spring Hauling 2019.pdf, 3-19 Sludge Data.pdf, 4-19 Sludge Data.pdf

Cindy,
See if any of this makes sense.
Doug

Douglas Langworthy
Assistant Water Resources Supervisor
City of Cadillac
231-775-2841

ATTACHMENT NAME:

TVS Sludge.xlsx

ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): spreadsheet (XLSX)

3/19

RAW SLUDGE	V. RAW SLUDGE
T.S.	T.V.S.
3.26	68.18
3.27	80.28
3.26	77.27
3.22	78.26
3.16	77.53
3.07	78.69
3.31	76.12
3.10	77.42
2.91	76.67
3.09	77.42
3.57	73.97
3.13	74.58
2.74	78.69
3.27	78.46
3.20	78.74
3.21	78.26
3.02	78.33
3.04	76.19
3.15	75.81
3.03	76.21
2.99	79.03
65.99	1616.10
3.14	76.96

AVG

4/19

RAW SLUDGE	V. RAW SLUDGE
T.S.	T.V.S.
2.89	68.85
3.19	77.61
3.46	73.53
3.13	75.41
2.38	76.09
3.40	78.79
3.18	77.97
3.22	77.05
3.03	87.93
3.35	77.33
2.94	80.00
3.17	79.03
3.17	81.67
2.96	78.33
2.98	80.88
2.84	78.08
3.10	77.97
2.89	80.00
3.03	77.97
3.41	70.77
3.75	72.73
65.47	1627.99
3.12	77.52

AVG

ATTACHMENT NAME:

Lab Data Spring Hauling 2019.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

[illegible]

23.22
26.86
22.29
26.76
31.56
7.31

138.0 DT

ATTACHMENT NAME:

3-19 Sludge Data.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

SLUDGE DATA

MONTH

MARCH

YEAR

2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
<u>3/1</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14400</u>	<u>98</u>	<u>1 TO 2</u>	<u>40,565</u>	<u>FROM #2</u>	<u>5660</u>
						<u>2 TO</u>		<u>FROM #4</u>	<u>7200</u>
DAYS TOTALS									
<u>3/2</u>	<u>6:00</u>	<u>9:00</u>	<u>180</u>	<u>14,400</u>	<u>97</u>	<u>1 TO 2</u>		<u>FROM #2</u>	
						<u>2 TO</u>		<u>FROM #4</u>	<u>6000</u>
DAYS TOTALS									
<u>3/43</u>	<u>6:30</u>	<u>9:30</u>	<u>180</u>	<u>14,400</u>	<u>98°</u>	<u>1 TO 2</u>		<u>FROM #2</u>	<u>14,945</u>
						<u>2 TO</u>		<u>FROM #</u>	
DAYS TOTALS									
<u>3/54</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14,400</u>	<u>97°</u>	<u>1 TO 2</u>	<u>20,740</u>	<u>FROM #2</u>	<u>8540</u>
						<u>2 TO</u>		<u>FROM #4</u>	<u>5400</u>
DAYS TOTALS									
<u>3/65</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14,400</u>	<u>97°</u>	<u>1 TO 2</u>		<u>FROM #2</u>	<u>8540</u>
						<u>2 TO</u>		<u>FROM #4</u>	<u>6000</u>
DAYS TOTALS									
<u>3/76</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14,400</u>	<u>97°</u>	<u>1 TO 2</u>	<u>18,605</u>	<u>FROM #2</u>	<u>8235</u>
						<u>2 TO</u>		<u>FROM #4</u>	<u>1800</u>
DAYS TOTALS									
<u>3/87</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14,400</u>	<u>97°</u>	<u>1 TO 2</u>	<u>6405</u>	<u>FROM #2</u>	<u>7930</u>
						<u>2 TO</u>		<u>FROM #4</u>	<u>4200</u>
DAYS TOTALS									
<u>3/98</u>	<u>7:00</u>	<u>10:00</u>	<u>180</u>	<u>14,400</u>	<u>97°</u>	<u>1 TO 2</u>	<u>64,660</u>	<u>FROM #2</u>	<u>8</u>
						<u>2 TO 3</u>	<u>61,305</u>	<u>FROM #4</u>	<u>9000</u>
DAYS TOTALS									
<u>3/109</u>	<u>6:00</u>	<u>9:00</u>	<u>180</u>	<u>14400</u>	<u>97°</u>	<u>1 TO 2</u>		<u>FROM #2</u>	
						<u>2 TO</u>		<u>FROM #4</u>	<u>4800</u>
DAYS TOTALS									

COMMENTS:

SLUDGE DATA

MONTH March YEAR 2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
3/10	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	9150
DAYS TOTALS						2 TO		FROM #	
3/11	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	5490
DAYS TOTALS						2 TO		FROM #4	6000
3/12	7:00	10:00	180	14,400	97°	* 1 TO 2	51,240	FROM #2	10,980
DAYS TOTALS						* 2 TO 3	26,230	* FROM #4	4800
3/13	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	7930
DAYS TOTALS						2 TO		FROM #4	4800
3/14	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	16,470
DAYS TOTALS						2 TO		FROM #	
3/15	7:00	10:00	180	14,400	98°	* 1 TO 2	54,290	FROM #2	12,505
DAYS TOTALS						* 2 TO 3	34,465	* FROM #4	2400
3/16	6:00	9:00	180	14,400	98°	1 TO 2		FROM #2	
DAYS TOTALS						2 TO		FROM #4	2400
3/17	6:30	9:30	180	14,400	98°	1 TO 2		FROM #2	
DAYS TOTALS						2 TO		FROM #4	3600
3/18	7:00	10:00	180	14,400	96°	1 TO 2		FROM #2	13,420
DAYS TOTALS						2 TO		FROM #	

COMMENTS:

SLUDGE DATA

MONTH

march

YEAR

2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
3/19	7:00	10:00	180	14,400	98°	* 1 TO 2	39,650	FROM #2	12,840
DAYS TOTALS						2 TO		FROM #	
3/20	7:00	10:00	180	14,400	97°	1 TO 2		FROM #2	10,370
DAYS TOTALS						2 TO		* FROM #4	4800
3/21	7:00	10:00	180	14,400	97°	1 TO 2		FROM #2	14,335
DAYS TOTALS						2 TO		FROM #	
3/22	7:00	10:00	180	14,400	98°	* 1 TO 2	50,135	FROM #2	8,845
DAYS TOTALS						2 TO 3	27,755	FROM #4	4800
3/23	6:00	9:00	180	14,400	98	1 TO 2		FROM #2	2,100
DAYS TOTALS						2 TO		FROM #	
3/24	6:00	9:00	180	14,400	98	1 TO 2		FROM #2	8,900
DAYS TOTALS						2 TO		FROM #	
3/25	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	12,200
DAYS TOTALS						2 TO		FROM #	
3/26	7:00	10:00	180	14,400	98°	1 TO 2	46,970	FROM #2	14,640
DAYS TOTALS						2 TO		FROM #	
3/27	7:00	10:00	180	14,400	98°	1 TO 2		FROM #2	
DAYS TOTALS						2 TO		* FROM #4	6000

COMMENTS:

SLUDGE DATA

MONTH

March

YEAR

2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
3/28	7:00	10:00	180	14,400	98°	1 TO 2		* FROM #2	8235
						2 TO		* FROM #4	3000
DAYS TOTALS									
3/29	7:00	10:00	180	14400	98°	1 TO 2	39650	FROM #2	2440
						2 TO 3	34465	FROM #4	15000
DAYS TOTALS									
3/30	6:00	9:00	180	14400	97°	1 TO 2		FROM #2	7320
						2 TO		FROM #	
DAYS TOTALS									
3/31	6:30	9:30	180	14400	97°	1 TO 2		FROM #2	10980
						2 TO		FROM #	
DAYS TOTALS									
						1 TO 2		FROM #2	
						2 TO		FROM #	
DAYS TOTALS									
						1 TO 2		FROM #2	
						2 TO		FROM #	
DAYS TOTALS									
						1 TO 2		FROM #2	
						2 TO		FROM #	
DAYS TOTALS									
						1 TO 2		FROM #2	
						2 TO		FROM #	
DAYS TOTALS									
						1 TO 2		FROM #2	
						2 TO		FROM #	
DAYS TOTALS									

COMMENTS:

ATTACHMENT NAME:

4-19 Sludge Data.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

SLUDGE DATA

MONTH April YEAR 2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
4/1	7:00	10:00	180	14400	98°	1 TO 2 2 TO		FROM #2 FROM #	12810
DAYS TOTALS									
4/2	7:00	10:00	180	14400	98°	1 TO 2 2 TO	49715	FROM #2 FROM #	12810
DAYS TOTALS									
4/3	7:00	10:00	180	14400	98°	1 TO 2 2 TO		FROM #2 FROM #4	1525 15600
DAYS TOTALS									
4/4	7:00	10:00	180	14400	98°	1 TO 2 2 TO		FROM #2 FROM #4	11590 3600
DAYS TOTALS									
4/5	7:00	10:00	180	14400	98°	1 TO 2 2 TO 3	42700 23180	FROM #2 FROM #4	9150 6000
DAYS TOTALS									
4/6	6:00	9:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #4	1525 7200
DAYS TOTALS									
4/7	6:00	9:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #	8235
DAYS TOTALS									
4/8	7:00	10:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #	9150
DAYS TOTALS							54290		
4/9	7:00	9:30 10:00	150 180	12000	96°	1 TO 2 2 TO 3	48670 35990	FROM #2 FROM #	2400
DAYS TOTALS									

COMMENTS:

SLUDGE DATA

MONTH

April

YEAR

2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
4/10	6:00	9:00	180	14400	97°	1 TO 2		FROM #2	8540
DAYS TOTALS						2 TO		FROM #	
4/11	7:00	10:00	180	14400	97°	1 TO 2		FROM #2	7320
DAYS TOTALS						2 TO		FROM #	
4/12	7:00	10:00	180	14400	96°	1 TO 2	48190	FROM #2	10370
DAYS TOTALS						2 TO 3	31110	FROM #	
4/13	6:00	9:00	180	14400	96°	1 TO 2		FROM #2	
DAYS TOTALS						2 TO		FROM #	
4/14	6:30	9:30	180	14400	96°	1 TO 2		FROM #2	
DAYS TOTALS						2 TO		FROM #	
4/15	7:00	10:00	180	14400	93°	1 TO 2	41480	FROM #2	3355
DAYS TOTALS						2 TO 3	40565	FROM #	
4/16	7:00	10:00	180	14400	94°	1 TO 2		FROM #2	610
DAYS TOTALS						2 TO		FROM #4	7200 *
4/17	7:00	10:00	180	14400	93	1 TO 2		FROM #2	6710
DAYS TOTALS						2 TO		FROM #4	7200
4/18	7:00	10:00	180	14400	94°	1 TO 2	54290	FROM #2	14640
DAYS TOTALS						2 TO 3	25010	FROM #2	

COMMENTS:

SLUDGE DATA

MONTH April YEAR 2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
4/19	6:30	9:30	180	14400	96	1 TO 2		FROM #2	
						2 TO		FROM #4	10,800
DAYS TOTALS									
4/20	6:00	9:00	180	14400	98	1 TO 2		FROM #2	4880
						2 TO		FROM #4	10,800
DAYS TOTALS									
4/21	7:00	10:00	180	14400	98	1 TO 2		FROM #2	4200
						2 TO		FROM #	
DAYS TOTALS									
4/22	7:00	10:00	180	14400	98°	1 TO 2		FROM #2	13420
						2 TO		FROM #3	
DAYS TOTALS									
4/23	7:00	10:00	180	14400	97°	1 TO 2		FROM #2	10980
						2 TO		FROM #4	
DAYS TOTALS									
4/24	7:00	10:00	180	14400	96°	1 TO 2	50935	FROM #2	6100
						2 TO 3	17690	FROM #4	6000
DAYS TOTALS									
4/25	7:00	10:00	180	14400	97°	1 TO 2		FROM #2	1830
						2 TO		FROM #4	12000
DAYS TOTALS									
4/26	6:00	9:00	180	14400	97°	1 TO 2	48,190	FROM #2	7015
						2 TO 3	21960	FROM #4	12,000
DAYS TOTALS									
4/27	6:00	9:00	180	14400	97°	1 TO 2		FROM #2	
						2 TO		FROM #4	10,800
DAYS TOTALS									

COMMENTS:

SLUDGE DATA

MONTH April YEAR 2019

RAW SLUDGE						TRANS. SLUDGE		SUPERNATANT	
DATE	TIME ON	TIME OFF	MIN PUMPED	GALS. PUMPED	SL. TEMP.	HOW PUMPED	GALS. TRANS.	DIGST. #2/#3/#4	GALS. DRAWN
4/28	6:00	9:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #	5490
DAYS TOTALS									
4/29	7:00	10:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #	9150
DAYS TOTALS									
4/30	7:00	10:00	180	14400	97°	1 TO 2 2 TO		FROM #2 FROM #4	7930 4800
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									
						1 TO 2 2 TO		FROM #2 FROM #	
DAYS TOTALS									

COMMENTS: